

Comparing children to LMs on a single sigmoid-slope axis

A derivation of the Chang & Bergen bridge

Standard Model 2 working note

May 9, 2026

1 The claim

Children’s per-word acquisition curves and large language models’ per-word acquisition curves both have a sigmoidal form in $\log(\text{experience})$. They live on the same axis. The *slope* of that sigmoid is the natural object to compare across the two populations.

This note derives the slope from each side, lines them up in matched units, and reports the headline numbers we use in the deck.

2 Children’s side: per-child sigmoid slope is κ_i

2.1 Linear predictor

In our model (slides 9–12), for child i , word j , age t :

$$\eta_{ij}(t) = \log r_i + \log \alpha_i + \kappa_i \log t - \delta_j, \quad P(\text{produces}_{ij} \mid t) = \sigma(\eta_{ij}(t)), \quad (1)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the standard logistic. Per-child parameters: input rate r_i , efficiency α_i , scaling exponent κ_i . Per-word: difficulty δ_j .

2.2 Per-child slope

Differentiate η_{ij} with respect to $\log t$, holding (i, j) fixed:

$$\frac{\partial \eta_{ij}}{\partial \log t} = \kappa_i. \quad (2)$$

Therefore $\text{logit}P(\text{produces}_{ij} \mid t)$ is a *linear* function of $\log t$ with slope exactly κ_i . The per-child sigmoid slope on $\log t$ is κ_i — by construction.

2.3 Population-marginalized slope

If instead we ask, “what slope does a researcher fit to *cross-sectional* Wordbank data for word w , marginalizing over the population of kids?”, the answer attenuates κ_{pop} .

For fixed word w and time t , define

$$X_i = \log r_i + \log \alpha_i + \kappa_i \log t,$$

so $P(\text{produces}_{iw} \mid t) = \sigma(X_i - \delta_w)$. With $(\log r_i, \log \alpha_i, \kappa_i)$ joint normal and (for simplicity) treating cross-correlations as zero,

$$\mathbb{E}[X_i] = \mu_r + \kappa_{\text{pop}} \log t, \quad (3)$$

$$\text{Var}[X_i] = \sigma_r^2 + \sigma_\alpha^2 + \sigma_\kappa^2 \log^2 t = \sigma_\xi^2 + \sigma_\kappa^2 \log^2 t =: \sigma_X^2(t). \quad (4)$$

The cross-sectional acquisition probability is

$$P_{\text{pop}}(t, w) = \mathbb{E}_i[\sigma(X_i - \delta_w)].$$

Using the standard logistic–Gaussian convolution approximation with $a^2 = \pi^2/8 \approx 1.234$ (Dau-nizeau, 2017),

$$P_{\text{pop}}(t, w) \approx \sigma\left(\frac{\mu_r + \kappa_{\text{pop}} \log t - \delta_w}{\sqrt{1 + a^2 \sigma_X^2(t)}}\right). \quad (5)$$

Differentiating with respect to $\log t$ at the midpoint $\log t_{50\%}$ where the numerator vanishes:

$$\boxed{\left. \frac{\partial \text{logit} P_{\text{pop}}}{\partial \log t} \right|_{t_{50\%}}} = \frac{\kappa_{\text{pop}}}{\sqrt{1 + a^2 \sigma_X^2(t_{50\%})}}. \quad (6)$$

For our M_best on English: $\kappa_{\text{pop}} = 10.4$, $\sigma_\xi^2 = 1.80^2 \approx 3.24$, $\sigma_\kappa^2 = 3.47^2 \approx 12.0$. Plugging in:

- At $t_{50\%} = a_0$ ($\log t = 0$): $\sigma_X^2 = 3.24$, slope $\approx 10.4/\sqrt{1 + 1.234 \cdot 3.24} = 10.4/2.24 \approx 4.6$.
- At $\log t = 0.4$ (about 28 mo if $a_0 = 19$): $\sigma_X^2 \approx 3.24 + 0.16 \cdot 12 = 5.16$, slope $\approx 10.4/\sqrt{7.37} \approx 3.83$.

So a researcher fitting a single sigmoid to Wordbank cross-sectional production data per word would observe *population-marginalized* slopes of about 4 logits per natural-log-time unit. The per-child slope is much higher ($\kappa_i \sim 10.4 \pm 3.5$); the cross-sectional fit averages over kids and the slope shrinks.

2.4 Which slope is the right LLM comparand?

Both kid-side slopes have meaningful comparands on the LM side. Chang & Bergen fit *per-LM*, *per-word* sigmoids: there’s no “population of kids” equivalent because each LM is a fixed deterministic learner. The LM analog of κ_i is therefore the fitted slope $1/\text{scale}_w$ for one (LM, word) pair — same “instance” (a single deterministic learner on a single word). That’s the comparison we make in the deck. The population-marginalized slope is what one would compare to a cross-LM-architecture pooled fit.

3 LM side: Chang & Bergen sigmoid fit

3.1 Their parameterization

For each (LM, word) pair, Chang & Bergen (2022) fit per-word mean surprisal S_w during training as a function of $x = \log_{10}(\text{steps})$, using the four-parameter logistic

$$S_w(x) = \text{ParamLower}_w + \frac{\text{ParamUpper}_w - \text{ParamLower}_w}{1 + \exp((x - \text{ParamXmid}_w)/\text{ParamScale}_w)}. \quad (7)$$

ParamUpper_w is the asymptote at $x \rightarrow -\infty$ (high surprisal at training start), ParamLower_w is the asymptote at $x \rightarrow +\infty$ (low surprisal after learning). ParamXmid_w is the x at the curve’s midpoint, and ParamScale_w is the scale parameter.

3.2 Translating to “proportion learned”

Define proportion-learned as the fractional surprisal reduction:

$$P_w^{\text{LM}}(x) := \frac{\text{ParamUpper}_w - S_w(x)}{\text{ParamUpper}_w - \text{ParamLower}_w} = 1 - \frac{1}{1 + \exp((x - \text{ParamXmid}_w)/\text{ParamScale}_w)} = \sigma\left(\frac{x - \text{ParamXmid}_w}{\text{ParamScale}_w}\right) \quad (8)$$

So P_w^{LM} is a sigmoid in $x = \log_{10}(\text{steps})$ with slope on logit:

$$\frac{\partial \text{logit} P_w^{\text{LM}}}{\partial x} = \frac{1}{\text{ParamScale}_w}. \quad (9)$$

3.3 Unit conversion from $\log_{10}$ to natural log

Let $y = \ln(\text{steps})$. Since $x = y / \ln 10$,

$$\boxed{\frac{\partial \text{logit} P_w^{\text{LM}}}{\partial y} = \frac{1}{\text{ParamScale}_w} \cdot \frac{1}{\ln 10} \approx \frac{0.434}{\text{ParamScale}_w}}. \quad (10)$$

Per-word, per-LM sigmoid slope on *natural-log* training experience.

4 Direct comparison

Equations (2) and (10) are on identical units: “logit per natural-log unit of experience.” Reading from our M_best fits and Chang & Bergen’s processed sigmoid fits:

	median slope	IQR
Children, per-child κ_i (English M_best)	10.3	[8.0, 12.6]
Children, per-child κ_i (Norwegian M_best)	12.5	[10.0, 14.9]
LMS, BERT per-word	0.76	[0.42, 1.32]
LMS, BiLSTM per-word	0.87	[0.53, 1.52]
LMS, GPT-2 per-word	0.81	[0.45, 1.54]
LMS, LSTM per-word	0.96	[0.44, 1.90]

Headline. Children are 10–15 $\times$ steeper on the same sigmoid-slope axis. LMS cluster at $\kappa \approx 1$, the unit accumulator boundary; children sit in a wide distribution around $\kappa \approx 10$.

5 Caveats

1. **Per-instance vs. per-architecture, but different “instance.”** Kids’ slopes vary across *children*; LM slopes vary across *words* (within an LM, each word gets its own $1/\text{scale}_w$). A single LM is a deterministic learner, so there’s no population-of-LM-instances analog of σ_κ within architecture — variation is across the word set, not across replicates.

2. **Different experience units.** The $\log t$ axis on the kid side is log-months-of-life. The x axis on the LM side is log-training-steps; for batch size 128 and average sentence length ~ 25 , that’s roughly $\log(\text{training-tokens}/3200)$. On the log axis, absolute alignment doesn’t matter for slope; relative changes in experience do.
3. **Cross-sectional Wordbank fit would give a smaller slope.** A researcher who fit a sigmoid to Wordbank’s per-word production-vs-age data (one curve per word, marginalizing over kids) would observe ~ 4 on the same axis, not 10. That’s the population-marginalized slope from equation (6). The $10\times$ headline is the *per-child* slope.
4. **LM training distribution is not CDS.** Chang & Bergen trained on BookCorpus + WikiText, which is adult written text, not child-directed speech. Per-word acquisition slopes there reflect distributional statistics of that corpus, not what kids hear. See the next section.

6 Forward direction: CHILDES-trained GPT (Feng et al. 2026)

Chang & Bergen’s comparison is a strong start but conflates two factors: (1) the structural difference in scaling between humans and LMs, and (2) the input distribution mismatch (adult-written-text vs. CDS). Feng et al. (2026) train GPT-style models on CHILDES, which matches the input distribution kids actually receive.

If their models retain training-step-by-step surprisal (or can be rerun with that logging), we can:

1. Fit Chang & Bergen-style sigmoids to CHILDES-trained GPT surprisal trajectories on *exactly the CDI vocabulary*.
2. Re-extract per-word slopes $1/\text{scale}_w$ on natural-log experience.
3. Replot Figure 18 of the deck (children-vs-LMs sigmoid-slope distributions) with these matched-input slopes.

The structural-difference hypothesis predicts the gap will largely remain: kids will still be $10\times$ steeper than LMs even when LM training is CDS-matched. If the gap shrinks substantially, the input-distribution explanation is doing more work than we currently attribute to it.

References

- [1] Chang, T. A., & Bergen, B. K. (2022). Word Acquisition in Neural Language Models. *Transactions of the Association for Computational Linguistics*, 10, 1–16.
- [2] Daunizeau, J. (2017). Semi-analytical approximations to statistical moments of sigmoid and softmax mappings of normal variables. *arXiv:1703.00091*.
- [3] Feng, S., et al. (2026). Child-directed-speech-trained language models. (Pending — to be filled in once we read the paper.)